
Build the Geometry In, Don't Hope It Emerges: A Reflection-Symmetry Audit of a Skeleton World Model

Anonymous Author(s)

Affiliation

Address

email

Abstract

The human body has an approximate left-right structure: reflecting a pose and exchanging anatomically corresponding left and right landmarks produces another valid pose representation. We use this structure to ask whether a self-supervised skeleton model learns a representation that behaves predictably under reflection. The study uses pose sequences derived from the Gait Abnormality in Video Dataset (GAVD) and a Skeleton Joint-Embedding Predictive Architecture (S-JEPA). Its target is a coordinate-derived comparison of left-side and right-side motion, not a diagnosis or validated clinical measurement.

The quality-controlled cohort contains 625 pose sequences from 93 source videos. Five outer cross-validation folds are created at the source-video level, so all sequences from one video remain together. In each fold, the encoder is trained from initialization on approximately 80% of the source videos and tested on the remaining approximately 20%. Four inner folds, built only from the outer-training sources, select the ridge read-out penalty. The outer-test sources are not used for model training, preprocessing, read-out fitting, or parameter selection. This protocol supports a narrow claim about post-development performance on held-out GAVD source videos; it does not establish generalization to unseen people or clinical validity. The registered paper run and governance reviews are incomplete, so this draft reports the method but no v2.1 empirical conclusion.

1 Evidence status

This document describes the registered laterality v2.1 protocol. The earlier transductive experiment evaluated an encoder on sequences that had also entered its representation training. That experiment generated useful hypotheses but cannot support a held-out-video claim. Its numerical results are superseded and intentionally omitted. Results may be added only after all registered folds, seeds, variants, evaluations, lineage checks, and governance requirements are complete.

2 Introduction

A world model learns useful structure by predicting or reconstructing aspects of its input without relying on the final task label. For a model of an articulated body, geometry supplies a natural test of what the representation has learned. We focus on bilateral reflection: the transformation that flips a skeleton horizontally and exchanges every named left body landmark with its right partner.

The central question is deliberately limited. We ask whether a representation trained without the laterality target contains useful information about a signed left-versus-right motion contrast and

33 whether that information changes correctly when the input is mirrored. We separate three statements.
34 A mathematical wrapper can force a read-out to change sign. A trained encoder followed by a
35 probe can exhibit useful sign-changing behavior. The encoder tokens themselves can satisfy a direct
36 reflection-equivariance test. The first statement is a construction; the latter two require empirical
37 evidence.

38 Laterality v2.1 uses fold-local training. A fresh encoder is trained for each outer fold, and the corre-
39 sponding outer-test videos remain unavailable until evaluation. This replaces the prior transductive
40 design and changes both the supported claim and the required computation.

41 3 Concepts and terminology

42 A **pose sequence** is a series of video frames represented by estimated body-landmark coordinates
43 rather than by the original images. Each sequence contains 33 BlazePose landmarks, and each
44 landmark has horizontal, vertical, and depth coordinates. A **source video** is the original video from
45 which one or more pose sequences were extracted. Because several sequences can come from the
46 same video, splitting sequences independently could put closely related material in training and
47 testing.

48 An **encoder** converts the pose sequence into learned numeric features. A **read-out**, also called a
49 probe, is a simpler model fitted on those features to predict the laterality target. The encoder follows
50 a fixed, pre-registered training schedule and does not use an early-stopping validation set. The read-
51 out uses inner cross-validation to select its ridge penalty.

52 **Cross-validation** rotates which data are held out. An **outer fold** measures final held-out perfor-
53 mance. An **inner fold** makes a training-only choice without consulting the outer test set. **Stratifica-**
54 **tion** keeps the source counts for the dataset annotations reasonably balanced. These annotations are
55 used for balancing and a named confounding control; they are not diagnoses or primary prediction
56 labels.

57 4 Dataset inventory and cohort construction

58 GAVD is the Gait Abnormality in Video Dataset [Ranjan et al., 2025, GAVD project, 2026]. Its
59 official distribution supplies annotations and public video links rather than raw media files. This
60 study operates on derived pose archives and does not redistribute source videos or identity-bearing
61 frames.

62 The frozen inventory contains 666 annotation files describing 103 source videos. A matching pose
63 archive is available for 642 annotations; 24 annotations have no matching archive. Pre-specified
64 quality-control rules retain 625 pose sequences from 93 source videos and exclude 17 sequences.
65 Exclusion can occur when the target cannot be computed, too few selected landmarks are valid, or
66 too few complete four-frame model patches remain. Inclusion never depends on whether the target
67 is positive, negative, large, or small.

68 Every retained sequence carries provenance information: the source annotation, pose-archive finger-
69 print, extraction version, pose-model identity, and visibility threshold. A SHA-256 content digest
70 binds the cohort table and arrays together. Later stages reject content or provenance that does not
71 match the frozen protocol.

72 5 Coordinate-derived laterality target

73 The target compares movement on the left and right sides of the pose representation. It uses five
74 anatomical pairs: shoulders, knees, ankles, heels, and foot-index landmarks. The hips define the
75 pelvis-centered coordinate system but are excluded from the target because pelvis centering makes
76 their velocity magnitudes algebraically identical.

77 For each pair, the calculation keeps only **paired-valid transitions**: both landmarks must be observed
78 at both ends of the same frame transition. Let $m_{L,k}$ and $m_{R,k}$ denote the median left and right speeds

79 for pair k . The normalized contrast is

$$c_k = \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}, \quad y = \text{mean}_k c_k.$$

80 The small constant ϵ prevents division by zero. Every pair must provide at least eight shared valid
81 transitions, and all five pairs must be usable. Positive and negative values indicate relatively greater
82 left-side and right-side motion, respectively, under this formula. These are mathematical meanings,
83 not clinical interpretations.

84 Short missing stretches may be interpolated for encoder input, but interpolated coordinates never
85 define the target. Invalid coordinates have a validity mask and cannot contribute as numeric place-
86 holders. Automated tests confirm that changing invalid placeholders leaves the target unchanged,
87 mirroring negates the target, and mirroring twice restores the input.

88 6 Training, validation, and testing design

89 6.1 Why the source video is the splitting unit

90 All sequences derived from one source video are assigned together. Mirrored inputs and training
91 augmentations inherit that assignment. No clip, mirror, or augmentation from an outer-test source
92 can enter the corresponding outer-training set.

93 GAVD does not supply a persistent person identifier, and this project does not infer identity. Two
94 different source videos could contain the same unidentified person. The split therefore supports
95 held-out-source-video evaluation, not held-out-person evaluation.

96 6.2 Five outer folds

97 The 93 eligible source videos are divided into five stratified outer folds. Approximately four fifths
98 of the sources train the encoder and fit the final read-out, while approximately one fifth provide the
99 untouched test set. Sequence counts vary because source videos contribute different numbers of
100 sequences.

Table 1: Registered outer folds. Splitting and balancing operate on source videos, not sequences.

Fold	Train sources	Test sources	Train sequences	Test sequences
0	74	19	436	189
1	74	19	443	182
2	74	19	553	72
3	75	18	548	77
4	75	18	520	105

101 This is not one permanent 80/20 split. Each source serves as an outer-test source exactly once and
102 as an outer-training source in the other four folds. Combining out-of-fold predictions gives one
103 held-out prediction for every eligible sequence.

104 6.3 Four inner validation folds

105 Each outer-training set is divided into four inner folds. In one inner round, 55–57 sources fit a
106 candidate read-out and 18–19 sources validate its ridge penalty. The validation role rotates until
107 every outer-training source has served as inner validation exactly once.

108 Because sources contribute different numbers of sequences, an inner fitting partition contains 266–
109 450 sequences and an inner validation partition contains 80–197 sequences. These are rotating
110 read-out partitions, not additional encoder-training or final-test sets. Exact memberships are stored
111 in the split artifact.

112 These inner folds do not select an encoder checkpoint, stop encoder training, or change the registered
113 encoder schedule. After penalty selection, the read-out is fitted again on all 74–75 outer-training
114 sources and applied to the corresponding 18–19 outer-test sources. Feature scaling, target scale,
115 neutral-band calibration, and other fitted preprocessing use outer-training data only.

116 7 Fold-local representation training

117 The representation model is an S-JEPA: a Skeleton Joint-Embedding Predictive Architecture. It
118 receives 64 time steps with 33 landmarks, groups time into four-frame patches, and uses a 96-
119 dimensional embedding, a four-layer encoder, a two-layer predictor, and four attention heads. Dur-
120 ing self-supervised training, portions of the sequence representation are masked and the model pre-
121 dicts the hidden latent content. The laterality target and dataset annotation do not enter this objective.

122 For each outer fold, optimization seed, and training variant, a new encoder starts from a recorded
123 seed-matched initialization and sees only that fold’s outer-training sources. Sampling first chooses
124 a source video uniformly and then one sequence from that source, preventing videos with many
125 sequences from dominating.

126 The paper profile registers five seeds, numbered 42 through 46, and two variants. The `vanilla`
127 variant never reflects a training sample. The `reflection_augmented` variant reflects a sample
128 with probability 0.5. Within a seed, the variants share initialization, source draws, masks, geometric
129 views, optimizer schedule, and update count; reflection decisions use a separate random stream.
130 The complete design contains 5 outer folds $\times$ 5 seeds $\times$ 2 variants, or 50 fold-local encoders, each
131 following the fixed 300-epoch schedule.

132 8 Held-out evaluation

133 For a given fold and seed, feature extraction uses the encoder trained without that fold’s test sources.
134 The primary lane fits a free linear ridge read-out to the learned representation. Comparison lanes in-
135 clude an identically structured untrained encoder, a target-component self-consistency oracle, pooled
136 and missingness features, constructed one-pass and two-pass antisymmetric read-outs, and direct
137 encoder-token reflection tests. The oracle verifies that saved pair contrasts reconstruct the target and
138 is not a learned baseline.

139 The principal performance measure is source-balanced R^2 , which gives every source video equal
140 total weight. Mean absolute error, source-level sign accuracy, and mirror errors are secondary mea-
141 sures. Each checkpoint is evaluated separately. Metrics and squared mirror errors are computed
142 within a checkpoint before registered seeds are averaged; signed errors cannot cancel across seeds.

143 An analytically antisymmetric wrapper and an empirically equivariant representation are different
144 claims. Exact output oddness proves the wrapper’s construction. Useful learned content additionally
145 requires positive held-out predictive utility and improvement over the paired untrained initialization.
146 A strict representation claim separately requires low direct token-equivariance error and improve-
147 ment over initialization.

148 9 Uncertainty and decision rules

149 Optimization-seed variation and source-sampling uncertainty are kept separate. Five seeds show
150 how training changes with registered randomized choices. Separately, 2,000 bootstrap repetitions
151 resample source videos while preserving all sequences from a source together. The resulting in-
152 tervals are conditional on the fitted cross-validation pipeline; they do not imply that the observed
153 videos are a random sample of all people or recording conditions.

154 Decision thresholds are frozen in the protocol. Native probe behavior requires useful absolute held-
155 out prediction, sufficiently low mirror error, and improvement over the paired untrained encoder.
156 Strict representation behavior requires sufficiently low token-equivariance error and a favorable
157 learned-minus-initialized comparison. A null or adverse result remains reportable and does not
158 authorize post-result retuning.

159 10 Evidence status and reporting boundary

160 The v2.1 cohort and source split have been audited, but the registered paper computation and held-
161 out evaluation are incomplete. This document therefore reports no v2.1 performance number or
162 empirical conclusion. Older transductive estimates answer a different question and must not be
163 inserted into this protocol’s tables, abstract, figures, or conclusion.

Every eventual number must trace to artifacts carrying matching protocol, context, cohort, split, checkpoint, and implementation digests. A final table must include all five outer folds and all five registered seeds for the relevant variant. Synthetic smoke output tests software plumbing and is not empirical evidence.

11 Limitations

The design prevents source-video overlap but cannot guarantee that people are distinct across videos. The supported claim is post-development, within-GAVD cross-validated performance on held-out source videos. It is not a claim about unseen people, another dataset, clinical validity, prevalence, diagnosis, or treatment.

The target captures one signed left-versus-right motion contrast, not the full kinematic structure of gait, and it is not a validated clinical index. Learning claims remain conditional on the BlazePose schema, pose extraction, preprocessing, architecture, registered seeds, and measured controls. Unmeasured acquisition and missingness effects may remain.

12 Ethics and data use

Public availability is not authorization for research use or redistribution. The suite stores no raw videos or identity-bearing frames, but derived poses, source identifiers, embeddings, predictions, and checkpoints may remain linkable or sensitive. The institutional ethics determination, data-use review, and derived-pose release review are unresolved. Submission and artifact release remain blocked until all three reviews carry genuine dated internal references.

13 Reproducibility sequence

Notebook 00 freezes the protocol and reports governance. Notebook 01 audits cohort construction and the target. Notebook 02 creates source-level outer and inner folds. Notebook 03 trains fold-local encoders. Notebook 04 fits read-outs and produces held-out predictions. Notebook 05 aggregates registered checkpoints and uncertainty. Notebook 06 provides a separate gate for a possible external dataset; the GAVD review cannot authorize that dataset.

The split assignments are stored in `artifacts/paper/protocol_<digest>/splits/source_splits.json`. Registered rules are stored in `config/protocol.json` and summarized in `PROTOCOL.md` and `RUNBOOK.md`.

References

- GAVD project. Gait abnormality video dataset: Data access and responsible-use notes. GitHub repository, 2026. URL <https://github.com/Rahmyyy/GAVD>. Accessed: 2026-09-03.
- Rahm Ranjan, David Ahmed-Aristizabal, Mohammad Ali Armin, and Juno Kim. Computer vision for clinical gait analysis: A gait abnormality video dataset. *IEEE Access*, 13:45321–45339, 2025. doi: 10.1109/ACCESS.2025.3545787.